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Interests: innovation in generative AI, LLMs, natural language syntax/semantics, AI-driven NLP & parsing systems

Career Statement:

Driven technologist and skilled AI researcher with a unique blend of academic depth, technical engineering skills, and commercial intelligence. My foundation in theoretical linguistic model research (yielding 9 publications across 6 languages) complements hands-on experience engineering NLP parsing systems at Melange, where I've implemented in-house grammatical models and LLM prompting strategies for complex programmatic linguistic tasks in patent AI. This technical expertise is enhanced by my proven GTM record at Speechmatics, where I exceeded targets by 160% while serving as a strategic cross-functional technical sales liaison between Product, Marketing, and Finance teams. My unique focus resides not only in building and shipping innovative AI solutions for enterprise but also in their effective translation from technical capability into compelling business narratives that drive real-world adoption. I deliver full product lifecycles from bleeding-edge research conception through to practical implementations that showcase measurable, rapid time-to-value.

Melange Technologies • NLP Research Engineer • November 2024–present

- **AI System Architecture:** Designed and implemented core components of Melange's natural language parsing pipeline, creating production-quality NLP systems from theoretically-driven context-free grammar modelling and algorithmic design.
- **Research to Production:** Successfully bridged generative theoretical linguistic research with practical engineering, implementing novel grammar-based parsing solutions that outperform traditional approaches on complex syntactic structures.
- **Language Model Development:** Build and optimize custom parsing models using complex grammatical formalisms (CFGs, dependency grammars) that extract structured information from unstructured text with high precision.
- **Prompting Systems:** Develop sophisticated natural language prompting templates that enhance LLM performance on specialized linguistic tasks, demonstrating bias toward practical implementation over theoretical perfectionism.
- **Cross-functional Collaboration:** Regularly partner in strategic product & GTM initiatives to translate NLP insights into practical featurization for PMF, effectively communicating technical concepts to both clients & non-technical stakeholders.
- **Daily AI Integration:** Actively incorporate LLMs and AI tools to creatively accelerate development workflow.
- **Technical Documentation:** Write and maintain comprehensive documentation for dev tools, grammatical systems, and parsing pipelines, enabling creative longevity, broader team adoption, and continuous improvement of NLP capabilities.

Speechmatics USA • Strategic Business Development • February 2024–October 2024

- **Business Development:** Consistently exceeded all Business Development KPIs, achieving 160% of lead conversion quota, generating 1M in pipeline revenue through strategic prospecting, exceptional outreach, technical demos, and brand evangelism.
- **Partnerships:** Fostered partnerships with unicorn logos from cold outreach and on-the-ground relationship-building.
- **Cross-functional:** Serve as the representative Business Development liaison for the cross-functional Product team involved in strategically increasing usage and demand for specific non-English Language initiatives and maximizing the effectiveness of individual model uplifts and landings. Served as the representative Business Development liaison between Product Marketing and Finance to develop internal Competitive Intelligence and drive GTM outreach initiatives based on expert market analysis.
- **RevOps & Marketing:** Drove improvements to marketing & revenue operations, streamlining the efficacy of MQLs by highlighting inefficiencies in the data collection of web portal user input, and writing technically engaging public-facing copy.
- **Technical Hunger:** Regularly self-educated and interfaced with Product blogs, developer resources, and Sales Engineers.

UCLA Department of Linguistics • Graduate Ph.D Research Scientist • 2019–September 2023

- **Research vision:** Owned an individual research agenda deploying the latest in theoretical linguistics across 6 languages.
- **Financial success:** Exceeded all KPIs in negotiating >200% of competitive funding quota for individual and team projects.
- **Language Model Design & Development:** Hands-on computational linguistic R&D experience developing FSGs,

weighted CFGs, and other non-ANN models across all domains of natural language in Haskell and Python.

- **Data Collection:** Designed and oversaw elicitation sessions extracting 300+ hours of language data from 12 participants.
- **Data annotation:** Hands-on experience with data annotation (Praat) and analysis to aid in the development of theoretical phonological, syntactic, semantic, and pragmatic models of natural language and communication.
- **Statistical:** Hands-on experience performing statistical analyses of experimental data for linguistic research using R.
- **Innovation & knowledge sharing:** Regularly collaborated to produce novel discoveries in natural language science.
- **Presentation & publication:** Contributed to team success authoring 13 conference presentations, publishing 9 papers.
- **Global representation:** Traveled to 5 countries in Europe/Asia delivering technical presentations on behalf of the team.
- **Industry event organization:** Led conferences with 300+ attendees, multitasking speaker selection, website design, social media, and team management, building and maintaining strong relationships across international universities.
- **Technology adoption:** Stayed abreast of the latest developments in technological trends, producing research summaries for departmental strategic planning, sharing knowledge, and driving innovation independently and as a key team player.

Harvard Journal of East Asian Linguistics (JEAL) • Invited Reviewer • 2022-2023

- **Expert authority:** Invited to perform peer review of journal submissions in the areas of Chinese theoretical syntax.

USC North American Summer School in Language, Logic, and Information • Career Mentor • 2022

- **Dedicated mentorship:** Invited to consult, providing valuable one-on-one guidance, expert career training, and mentorship services, nurturing the educational ambitions of aspiring undergraduate researchers in the natural language sciences.

UCLA Department of Linguistics • Primary Instructor and Teaching Assistant • 2020-2022

- **Exceptional student success:** Achieved a 98% pass rate as the primary instructor for UCLA's undergraduate linguistics.
- **Engaging curriculum:** Designed and delivered comprehensive coursework for over 750 students, fostering excellency.
- **Effective leadership:** Navigated academic logistics by bridging communication between admin, faculty, and students.
- **Team management:** Led a team of assistants, ensuring their support and effective contribution to the learning process.

Education

- M.A., Generative Linguistics, 2023. University of California, Los Angeles (UCLA) Department of Linguistics.
- Ph.D coursework in mathematics, computer science, cognitive science, psychology, formal logic, and NLP.
- M.A., Theoretical Linguistics, 2019. Institute of Linguistics, National Tsing Hua University (NTHU).
- B.A., Cognitive Science and Mandarin Chinese, 2016. Marlboro College. Graduated with Highest Honors.

Proficiencies & Technical Experience

- *Programming & Version Control:* Python, Haskell, Lisp, Go, fish, zsh, Bash, Git, Docker, JavaScript, R, LaTeX
- *NLP Tools:* Stanza, spaCy, NLTK, Praat, ELAN, Jupyter
- *AI Integration:* LLM APIs (OpenAI, Anthropic), Cursor AI (VS Code), emacs + aidermacs
- *Data:* Linguistic Corpus Development, Data Annotation, Statistical Analysis, Experimental Research Design
- *Business Intelligence:* Technical Demo Design, CRM (Salesforce), Product Positioning, GTM Strategy
- *Languages:* Polish (fluent, heritage), Mandarin (fluent)
- *Soft:* Cross-functional Collaboration, Technical Documentation, Research Planning, Technical Presentations